AUTOMATED SMART IRRIGATION SYSTEM (ASIS)

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Abstract- A smart city boasts of services that aim to enhance the quality of life by not only providing the basic amenities, but also the efficiency and performance of systems and processes that ensure the economics and sustainability. A major emphasis is placed on environmental quality and habitable ambience within the city. Greenbelts not only improve the air quality but also lend an aesthetic appearance to the city. However, the maintenance and upkeep of the greenbelts is resource-intensive due to the manual nature of the work and inefficient irrigation methods. Conventionally, the city's greenbelts are watered manually by using pipelines fed through a mobile water tanker, which moves at a slow pace, while a worker directs the water pipe to the greenbelts. This process does not take into account the variations in the water requirements of different plants as well as the soil moisture levels and also the efficiency of irrigation and hence are wasteful. A smart solution can be a smart irrigation system that takes into account the water requirements, soil moisture levels, ambient environmental conditions, air moisture level and optimizes the water usage with minimal human intervention.

Keywords: Irrigation system, Internet of Things, soil moisture sensor, temperature sensor, crop quality, fertilizer, NPK sensor, ARIMA

1. INTRODUCTION

India is an agriculture based country where 70% of land comes under agriculture Agriculture and its allied activities is the main source of livelihood for approx 80% of the total rural population. In which almost 52% of the labor is employed for cultivation. Agriculture contributes to the country's GDP in between 14 – 15 %. As the population is increasing, simultaneously the requirement is rising, for that there is an immense need to modernize the conventional method of agriculture.

In agriculture it is big and complex problem in hand to solve, how much water is required for crops and in which interval through an automated system. A smart automated system can solve it using new and smart technologies like IoT, sensors and wireless technologies to operate in from remote location, and this is the main objective this paper.

This solution can solve big concern of farmer and other hand can save time, water and human resources. Solution will achieve this ability using different sensors (e.g. temperature, soil moisture sensor, air moisture sensor) and decision making will done by micro controllers. The sensed data received from sensors will be sending to server database using wireless transmission.

In crop cultivation, field crop needs to be managed in terms of pesticides, fertilizers, and irrigation for better growth of crops. Precision agriculture (PA) is the concept used. The sensed data such as temperature, humidity, fertilizer, and soil moisture can be provided as input by PA to decision support system in order to maximize the crop growth. Also the objective should be maximizing the crop growth with the optimized use of available resources and without affecting the environment.

2. LITERATURE REVIEW

For crop yield in agriculture predicting the amount and time to provide water plays a crucial role. Future planning can be done for the crop production, storage and their marketing to the market. Inclusion fo risk management techniques can be included for the future goals. To estimate production and watering strategies for a specific crop in advance two methods are used. Statistical methods(ARIMA) and Holt-Winter method is being used [9]. The second method is machine learning methods like (SupportVectorMachine and artificial neural network[10]). Agriculture in India is hugely dependent on rainfall, especially in summer season so create such system is utmost important to save water resources and offer more yield. Artificial neural network is used in system to process information data collected by sensors, to predict production yield of selected crop. This information includes various parameters like type of soil, temperature (average temperature), rainfall in selected area, and humidity. In this paper evolution has been done as how these crops related parameters affect crop growth and yield [11]. It is observed that different atmospheric parameters, soil type, soil composition and required soil moisture can impact on production rate of crop. The technique discussed in this paper also predicts suitable crop production rate in advance. Artificial neural network is used as an tool for modeling and prediction of crop yield [5]. Archana et al. (2016) presented a paper in which the humidity and soil moisture sensors are being studied. These sensors were placed in the plant's root. Based on those values, collected from the sensor, the microcontroller was used to control the supply of water to the field. This system was a little drawback as it fails to intimate the farmer about the field status [1]. V. R. Balaji et al projected a paper in which states that the system can derive power from sunlight through photovoltaic cells. This system has least dependency on electricity. The soil moisture sensor has been used and based on the sensed values PIC microcontroller is used to ON/OFF the motor pump. Weather forecasting is not included in this system [3]. Sonali D. Gainwar et al. proposed a paper, which states that in soil parameters such as pH, humidity, moisture and temperature are collected. This system is completely automated which turns the motor pump ON/OFF as per the level of moisture in the soil. The current field status is not downcast to the farmer[2]. R.Subalakshmi (2016) proposed a paper to make irrigation system more simpler. According to her paper, the complexities involved in irrigation can be tackled with automation system using microcontroller and GSM. Based on the sensed values from soil moisture, temperature and humidity sensors, the GSM sends message to the farmer based on these parameters when exceeded some fixed threshold value set in the program. Drawback is that the nutrient content in the soil is not determined [4]. Karan Kansara (2015) projected an automated irrigation system. In that they have studied using humidity and temperature sensors

to sense the soil conditions.Based on the results , microcontroller will control the water flow into the farming land. Farmer will be informed through Mobile devices. This system lacks in monitoring the nutrient content present in the soil [5]. G.Parameswaran and K.Sivaprasath (2016) proposed a smart drip irrigation system using IOT in which humidity, temperature and pH sensors are used. Irrigation status is updated to the server or local host using personal computer. In this work the farmer must have access to the internet in order to get the field condition report [6]. S.Reshma and B.A.Sarath (2016) proposed an IOT based automatic irrigation system using WSN.This system has various sensors to measure the soil parameters. This system was further connected to a GUI with the web to the user.This helps them to monitor and control the system from distant location. Weather monitoring was lacking as well [7]. Ahmad et al. (2001) analyzed water quality data using an ARIMA model.

3. SENSORS USED

This system is based on Nodes, which are responsible for sending data from sensors. Each sensor should be connected to this node. Node can be a cloud gateway or Wifi connection, or a serial connection. Sensors are required to be connected to a (WiFi based) microcontroller, further the microcontroller can read periodically the sensor and then it can send the sensor data to Raspberry pi may be by using protocols such as Message Queuing Telemetry Transport (MQTT), HTTP, TCP, UDP. However, Raspberry pi itself can read the data from that WiFi microcontroller periodically by hosting a server. There are many options available for choosing microcontroller such as ESP8266 boards (Nodemcu/Wemos d1 mini), or Arduino with ESP8266 or some other boards like Linkit smart 7688 or Onion Omega etc.

SOIL MOISTURE SENSOR

frequency domain technology is utilized to find the volumetric water constant(VMC) of the soil. This can be done by using sensor which measures these on the basis of dielectric constant parameters .

The above mentioned sensor consumes low power and has very high resolution. This increases its feature to make more accurate measurements (i.e. on hourly basis) over a long period of time with minimal battery usage. In addition, the sensors include high frequency oscillation as an inherent feature, which helps the sensor to record the soil moisture accurately with minimal salinity and textural effects.

HUMIDITY SENSOR

A humidity sensor (or hygrometer) provides the complete report on moisture and air temperature both.

SOIL NUTRIENT SENSOR

Fertilizer is added to soils to increase plant productivity. A common type of fertilizer is called NPK fertilizer because of its ingredients: nitrogen (N), phosphorus (P), and potassium (K). So to complete an automated irrigation system, monitoring nutrient content of soil is necessary to decide how much extra contents of these nutrients are to be added in the soil to increase crop fertility.

WATER DEPTH DETECTOR

Water depth sensor can measure the depth of the water level in the soil. Water depth sensor is very helpful in the irrigation process. This sensor will clearly help to record the water content level in the soil ,which ultimately help in monitoring the amount of water to be delivered to the fields during irrigation process. This will ensures the delivery of no excess water and also helpful in ensuring absence of shortage of water to the crops in the fields

4. METHODOLOGY

The proposed research work focuses on the use of effective IOT devices and efficient algorithms for refinement of sensed data. Following if the system flow (Figure 1):-

STEP 1: Installation of various sensors in the crop land.

STEP 2: Data generation from various sensors: can be on a daily basis at an interval of a fixed time duration.

STEP 3: Collected data is sent to the microcontroller / Raspberry Pi board.

STEP 4: Created time series dataset is then analysed using ARIMA for prediction. Final prediction on Water supply is made by combined result of real time weather forecast data and sensor generated data.

STEP 5: Decision tree algorithm implemented to make ON/ OFF of the pump

IoT devices –The usage of IOT devices has tremendously increased in the farming process, as a result farming process has been accelerated and the crop production has increased. Now a day's modern types of IoT devices are used in smart farming effectively which enhances the functionalities of farming. The smart farm, embedded with IoT systems, can support a wide range of devices from diverse agricultural device manufacturers [8].

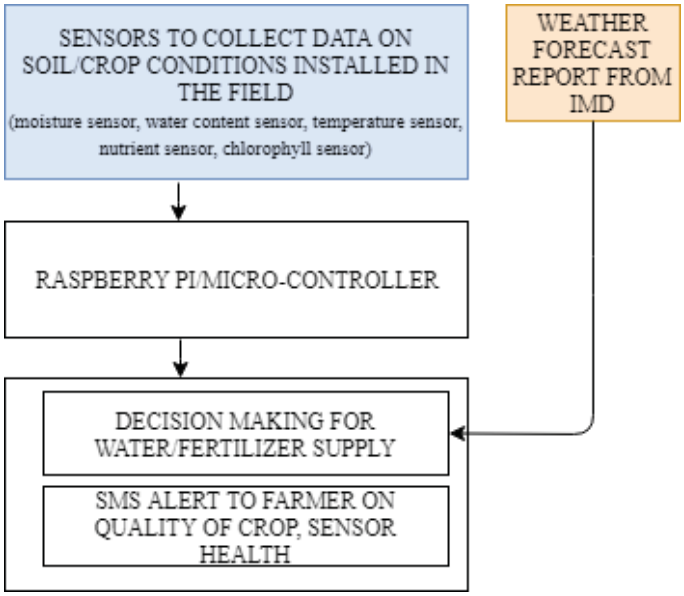


Figure 1. Flow diagram of the system

TIME SERIES ANALYSIS FOR PREDICTION

Time Series Analysis is widely used prediction model usually for predicting the weather conditions. However data generated from various sensors is time series data only, so this model can be built on any type of sensor data.

ARIMA

In general, an ARIMA model is characterized by the notation ARIMA (p, d, q) where p, d, q denote orders of auto-regression, integration (differencing) and moving average respectively. In ARIMA, time series is a linear function of past actual values and random shocks. It caters to a standard structures in time series data, and provides a simple yet powerful method for making time series forecasts.

- AR: Auto regression. A model that uses the dependent relationship between an observation and some number of lagged observations.
- I: Integrated. The use of differencing of raw observations in order to make the time series stationary.
- MA: Moving Average. A model that uses the dependency between an observation and a residual error from a moving average model applied to lag observations.

Each of these components is explicitly specified in the model as a parameter. A standard notation is used of ARIMA (p,d,q) where the parameters are substituted with integer values to quickly indicate the specific ARIMA model being used. The parameters of the ARIMA model are defined as follows:

- p: The number of lag observations included in the model, also called the lag order.
- d: The number of times that the raw observations are differenced, also called the degree of differencing.
- q: The size of the moving average window, also called the order of moving average.

Machine learning is the most important technology in knowledge mining. Due to the use of machine learning technique application pattern recognition in this field is simplified and accuracy of the result is improved. C4.5 decision tree learning algorithm is used in [7] to develop classifier extraction tool to categorize disease of rice crop in Egypt and surrounding area. Also till date, many tools are developed for prediction of crop growth or production rate in agriculture field.

5. WORKING OF ASIS

As represented in figure 2, an automated smart irrigation system is implemented, where sensors are installed in the field. The sensors used for this implementation are:- soil moisture sensor, NPK sensor, temperature sensor.

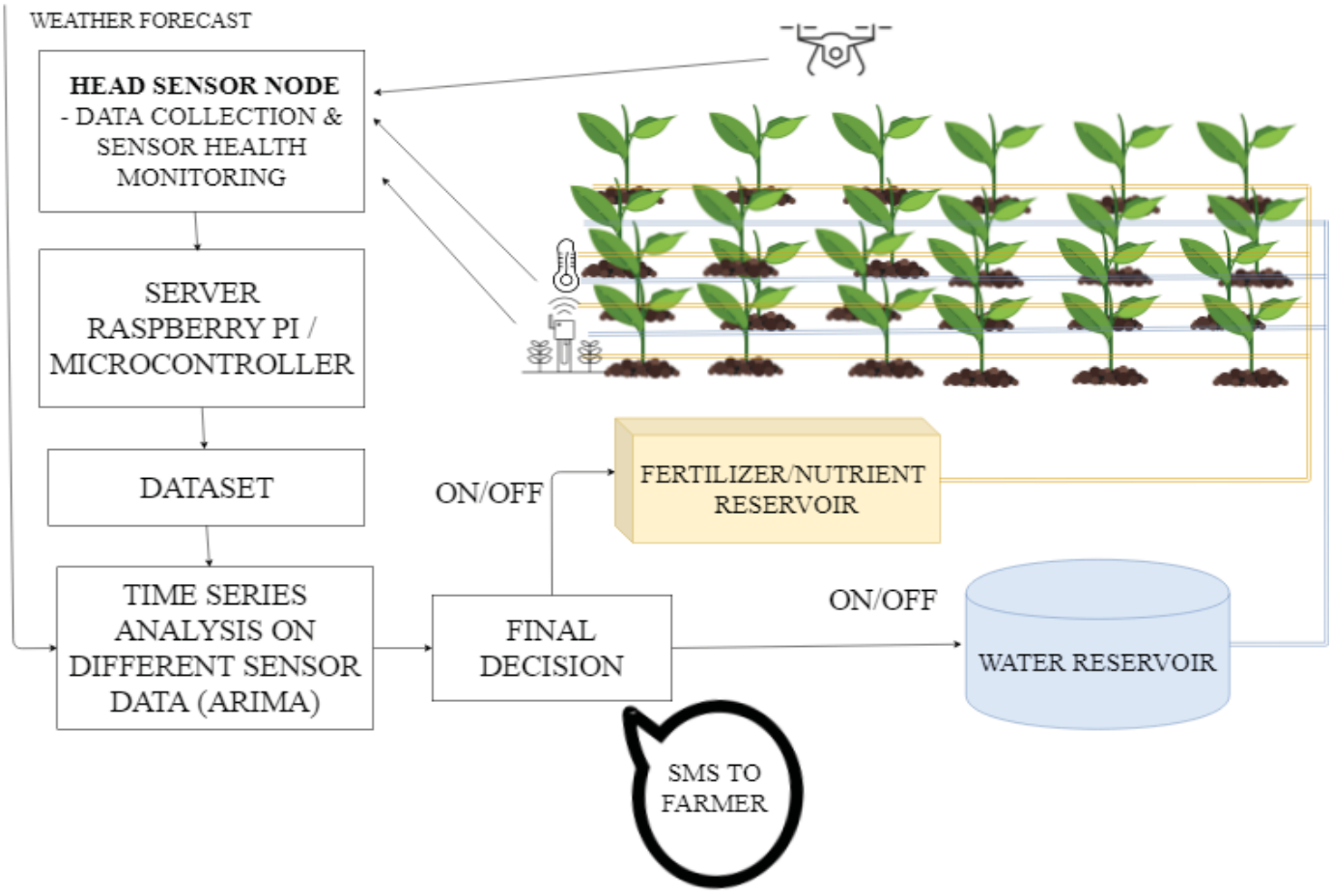


Figure 2. Working of Automated Smart Irrigation System

This agriculture drone is equipped with high density camera and it is supposed to capture images of crop throughout the field at different geographical locations. A part from these sensors, quality of crop is also being monitored through periodic patrolling on the field by drone.

Also when sensors are installed physically, their health monitoring is required for proper functioning of the automated system. For sensor health monitoring, all sensor nodes are connected to a head sensor node. This head sensor node collects data gathered by the sensor nodes and information about anomaly occurring at sensor node. It will alert the user on sensor node going down. The collected data is sent to the micro-controller board and dataset so generated is analysed using ARIMA for further prediction. The images collected from drone are also processed for their color component. Machine learning classification algorithm is used to identify bad quality images. For any indication of low quality of crop an SMS alert is sent to farmers. After all data has been processed, as per the prediction, pump for fertilizer and water will get opened or closed. For water pump, we have also included weather forecasting data for coming two days for better water management. The prediction from ARIMA model and forecast are mapped to know if water reservoir needs to be opened or not. Say if the weather forecast is of rainfall for the next coming day and ARIMA predicts requirement of water in fields, then water pump will be opened only if the soil moisture content is very below possibly leading to dried crop; else the water will not be pumped to the fields and the system will wait for expected next day rainfall.

5. CONCLUSION

Primary objective of this proposed self operated, smart automated irrigation solution is to enable framer save time and resource, save water resources, get more yield from crop and hence save cost.

Taking into consideration on all four parameters such as soil moisture, temperature, NPK soil component and crop quality through images has made this system more efficient and effective for automated irrigation. Moreover the inclusion of weather forecasting will help in water management.

REFERENCES

[1] Archana and Priya, ”Design andImplementation of Automatic Plant Watering System” presented at International Journal of Advanced Engineering and Global technology , vol-04, Issue-01 , Jan-2016

[2] Sonali.D.Gainwar and Dinesh.V.Rojatkar ,“Soil Parameters Monitoring with Automatic Irrigation System” presented at International Journal of Science, Engineering and Technology Research,,vol-04,Issue 11,Nov 2015

[3] R.Balaji and M.Sudha, “Solar Powered Auto Irrigation System” presented at International Journal of Emerging Technology in Computer Science and Electronics (IJETCSE), vol-20 Issue-2, Feb-2016.

[4] R.Subalakshmi and Anu Amal, “GSM Based Automated Irrigation using Sensors” presented at Special Issue published in International Journal of Trend in Research and Development (IJTRD), March-2016.

[5] Karan Kansara and Vishal Zaweri, ”Sensor Based Automated Irrigation System with IOT” presented at International Journal of Computer Science and Information Technologies, vol-06, 2015.

[6] El-Telbany M, Warda M, El-Borahy M. Mining the classification rules for egyptian rice diseases. Int Arab J Inf Technol 2006;3(4):303-7.

[7] Ryu M, Yun J, Miao T, Ahn IY, Choi SC, Kim J. Design and Implementation of a Connected Farm for Smart Farming System; 2015.

[8] Sujjaviriyasup T, Pitiruek K. Agricultural product fore-casting using machine learning approach

[9] Suhas M Patil, Sakkaravarthi R. Internet of Things based smart agriculture system using predictive analytics.

[10] Dahikar SS, Rode SV. Agricultural crop yield prediction using artificial neural network approach. Int J Innov Res Electr Electron Instrum Control Eng 2014;2(1):683-6.

[11] Singhal, Gaurav & Bansod, Babankumar & Mathew, Lini. (2018). Real time Crop health monitoring using Remote Sensing and ancillary information using GIS.